```
from google.colab import files
uploaded = files.upload()
```

Choose Files train.csv

- **train.csv**(text/csv) - 61194 bytes, last modified: 5/19/2025 - 100% done

Saving train.csv to train.csv

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load Titanic dataset
df = pd.read_csv("train.csv")

# Set plot style
sns.set(style="whitegrid")

# Show first 5 rows
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
				Futrelle, Mrs. Jacques Heath								

Next steps: Generate code with df View recommended plots New interactive sheet

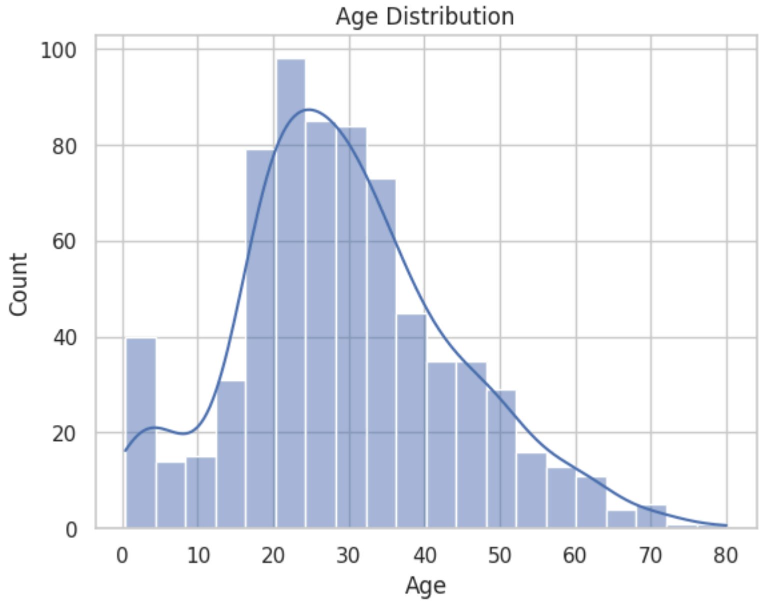
```
# Data info
df.info()

# Summary statistics
df.describe()

# Check for missing values
df.isnull().sum()

# Value counts for categorical variables
df['Survived'].value_counts()
df['Sex'].value_counts()
df['Pclass'].value_counts()
```

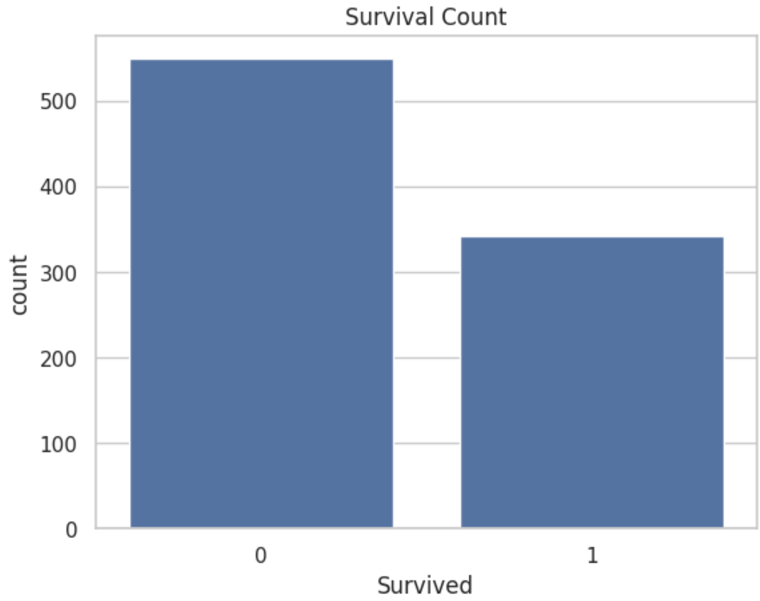
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

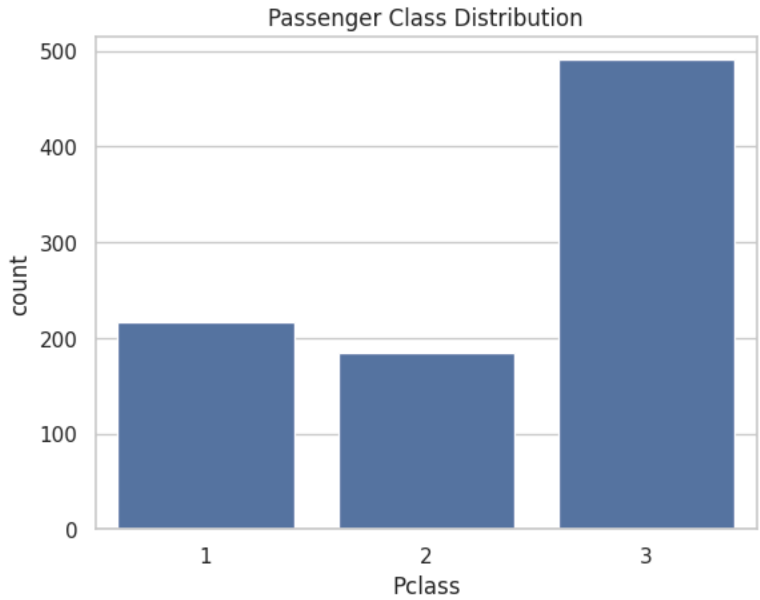
Pclass	count
3	491
1	216
2	184

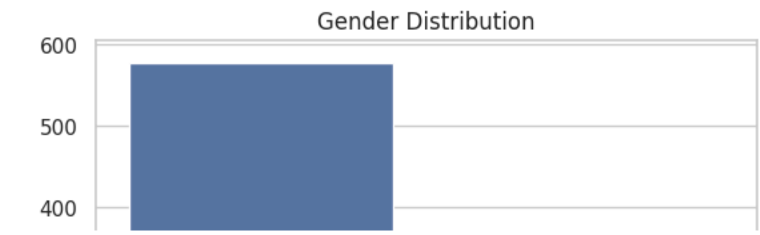
dtype: int64

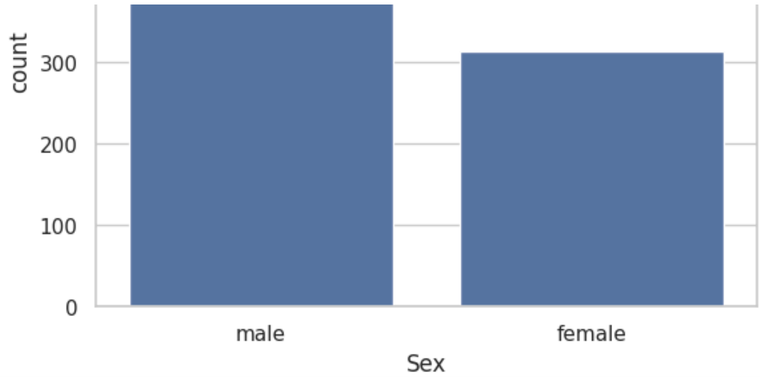
```
# Univariate Analysis

# Histogram of Age
sns.histplot(df['Age'].dropna(), kde=True)
plt.title('Age Distribution')
plt.show()

# Countplot for Survived
sns.countplot(x='Survived', data=df)
plt.title('Survival Count')
plt.show()

# Countplot for Passenger Class
sns.countplot(x='Pclass', data=df)
plt.title('Passenger Class Distribution')
plt.show()

# Gender distribution
sns.countplot(x='Sex', data=df)
plt.title('Gender Distribution')
plt.show()
```


Age Distribution
Count
0
20
40
60
80
100
0 10 20 30 40 50 60 70 80
Age


Survival Count
count
0
100
200
300
400
500
0 1
Survived


Passenger Class Distribution
count
0
100
200
300
400
500
1 2 3
Pclass

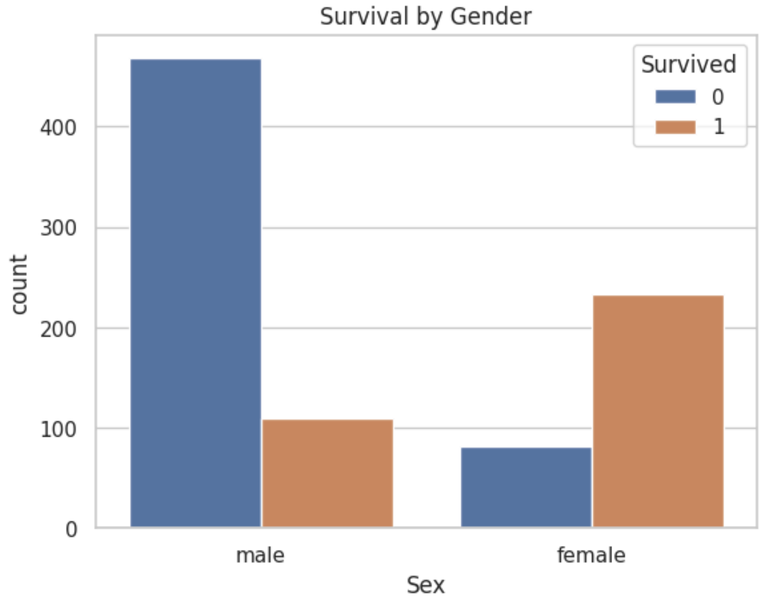

Gender Distribution
400
500
600

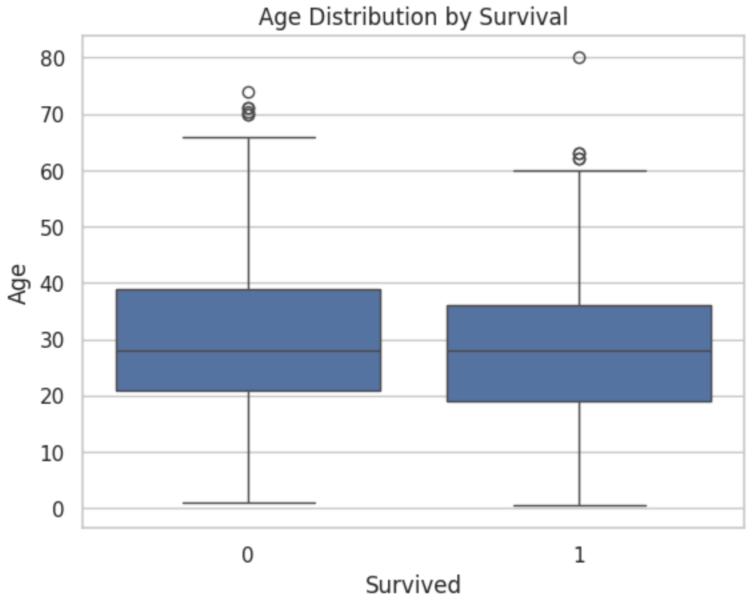


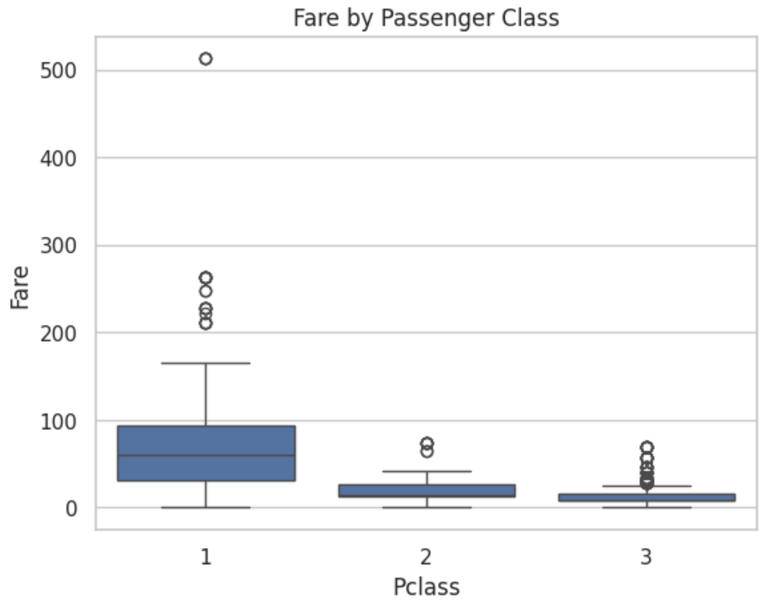
Here, the observation of each graph ;

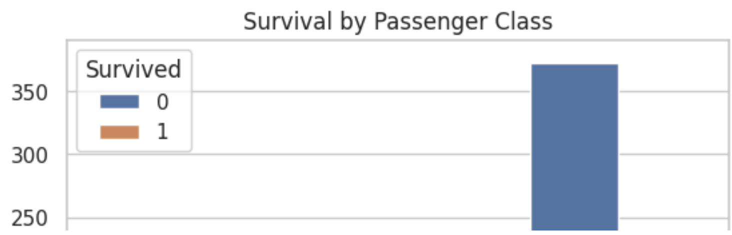
1. Age Distribution: The passenger age distribution is right-skewed with peaks in the younger to mid-adult range.
2. Survival Count: More passengers in the training data did not survive compared to those who did.
3. Passenger Class Distribution: The majority of passengers belonged to the 3rd class.
4. Gender Distribution: There were significantly more male passengers than female passengers

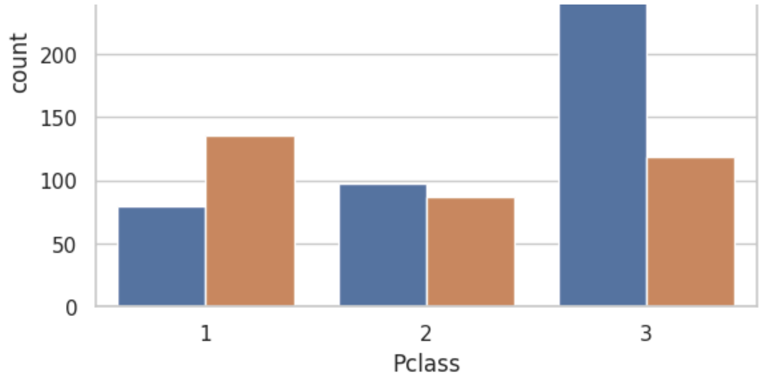
```
# Bivariate Analysis

# Survival by Sex
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Gender')
plt.show()

# Boxplot: Age vs Survived
sns.boxplot(x='Survived', y='Age', data=df)
plt.title('Age Distribution by Survival')
plt.show()

# Fare distribution by class
sns.boxplot(x='Pclass', y='Fare', data=df)
plt.title('Fare by Passenger Class')
plt.show()

# Survival by Pclass
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Passenger Class')
plt.show()

# Boxplot: Fare vs Survived
sns.boxplot(x='Survived', y='Fare', data=df)
plt.title('Fare Distribution by Survival')
plt.show()
```

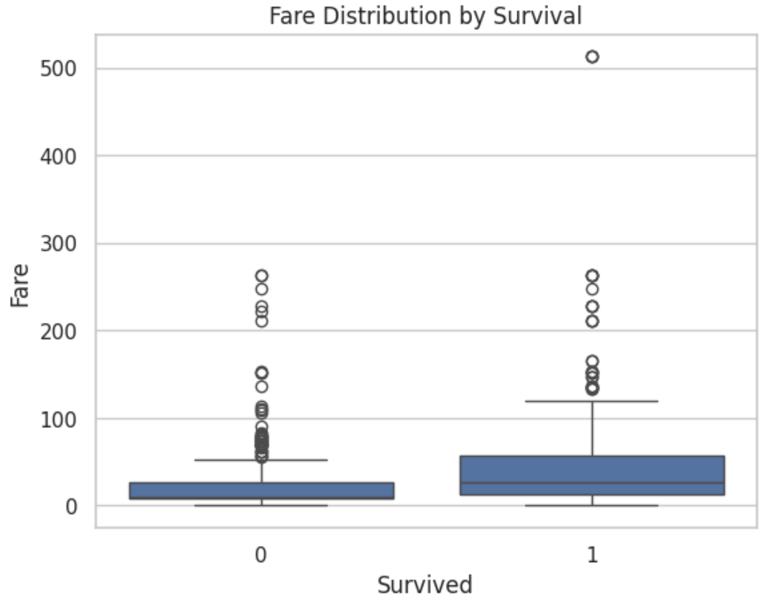

Survival by Gender
Survived
0
1
400
300
200
100
0
count
male
female
Sex


Age Distribution by Survival
80
70
60
50
40
30
20
10
0
Age
0
1
Survived


Fare by Passenger Class
500
400
300
200
100
0
Fare
1
2
3
Pclass


Survival by Passenger Class
Survived
0
1
350
300
250





Here, the observation of each graph ;

1. Survival by Gender: Females had a substantially higher survival rate than males.
2. Age Distribution by Survival: Survivors and non-survivors had similar median ages, but the non-survival group had more older outliers.
3. Fare by Passenger Class: Ticket fare increased with higher passenger class.
4. Survival by Passenger Class: First-class passengers had the highest survival rate, while third-class had the lowest.
5. Fare Distribution by Survival: Survivors generally paid higher fares than non-survivors.

```
# Heatmap of Correlations
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

```
 # Pairplot (optional)
sns.pairplot(df[['Survived', 'Pclass', 'Age', 'Fare']], hue='Survived')
plt.show()
```